

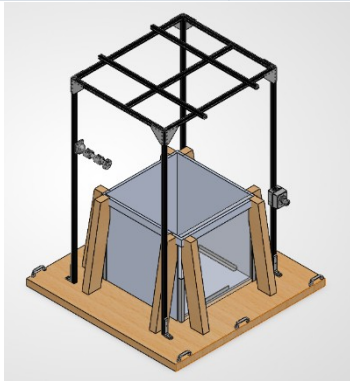
Matthew Walz — Engineering Portfolio

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B.S. Mechanical Engineering, Colorado School of Mines, expected May 2028 · GPA 3.9

Technical skills: SolidWorks design and simulation, Python, C++, LabVIEW, EES, OpenRocket, Mathcad Prime, Microsoft Office; Bambu 3D printing, Epilog laser cutting, machine-shop and hand-tool fabrication, sensor integration.

Project	Role	Highlight
Small Testbed Regolith Bin, RoSE Lab, Aug 2026–present	Subsystem lead	Designed a small lunar surface simulator for testing and observing regolith under load
Rocket Club, Tripoli Level 1, Spring 2026	Designer and builder	Designed, built, and flew an H-motor rocket to Level 1 certification
Tourniquet Feedback Device, Red Rocks CC, Spring 2025	Team lead, 5 students	Working blood-flow feedback prototype, 100% correct in validation tests

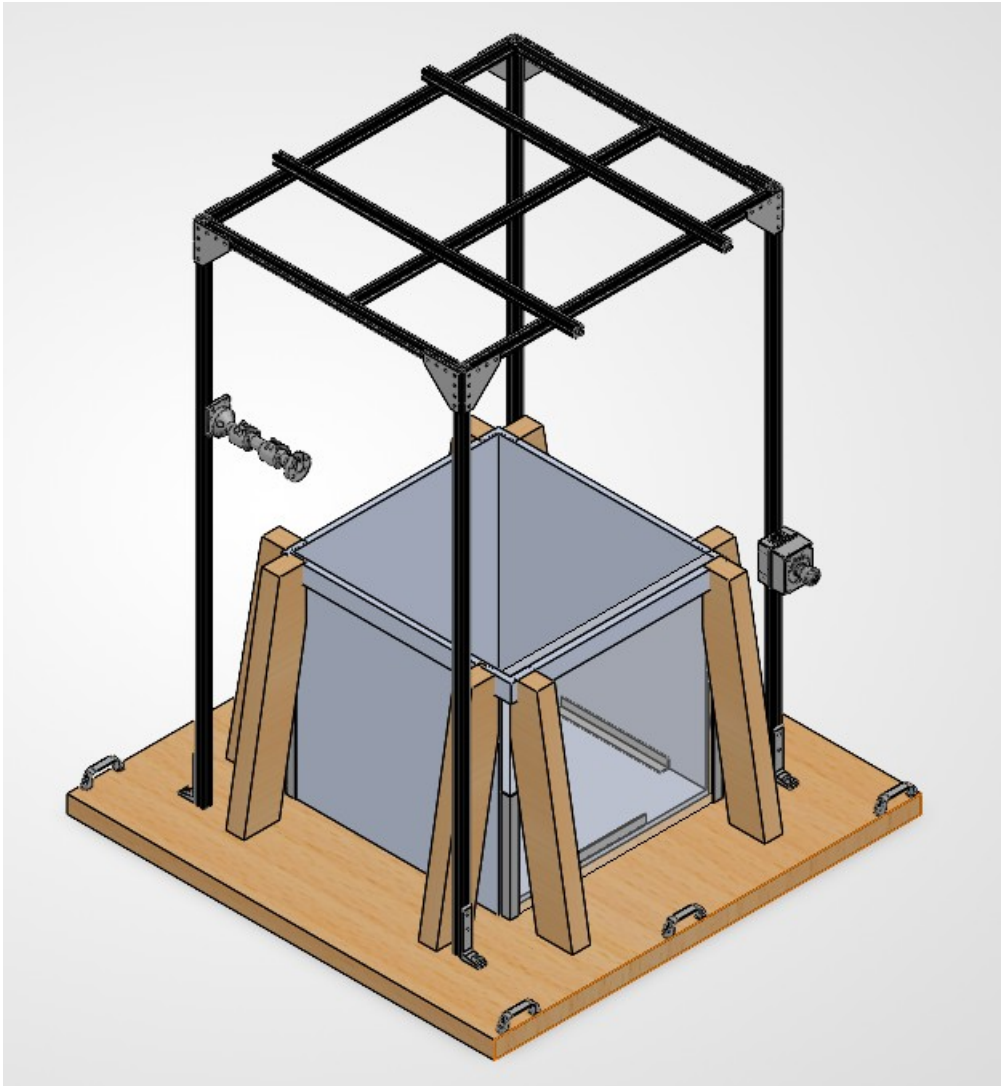


Small Testbed regolith bin, Level 1 rocket on the pad, and the tourniquet feedback device

Small Testbed Regolith Bin

Colorado School of Mines RoSE Lab • Bevameter project • Aug 2026–present • Subsystem lead

A 60 cm benchtop bin that holds compacted lunar regolith simulant while the Bevameter drives test plates into it, sized for an 18 kN bounding load. I own the design, analysis, fabrication, test procedures and assembly; presented it at the team's Critical Design Review.

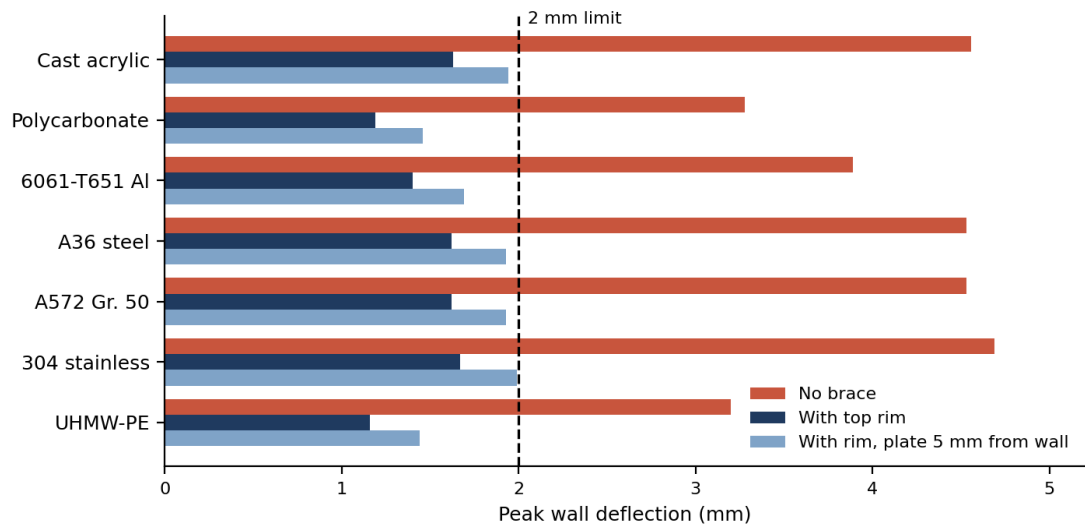


SolidWorks model: aluminum T-slot gantry over the regolith bin, with the top rim, corner braces and transport base

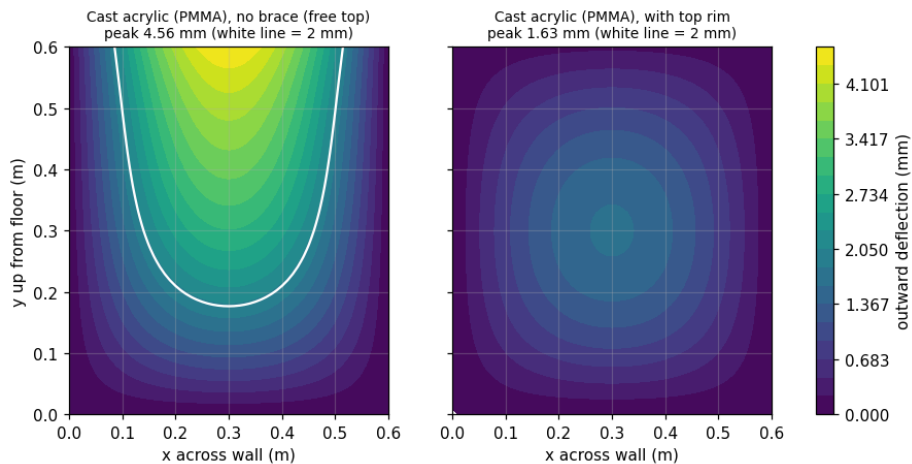
Key result	Value
Bounding load on frame and actuator mounts	18 kN
Wall materials meeting the 2 mm deflection limit	0 of 7 → 7 of 7 with the top rim
Selected walls	6061 aluminum, with a cast-acrylic viewing window
Selected frame	1 × 1 in aluminum T-slot extrusion

The key finding: an open-top bin needs a stiff rim

I modeled the compacted soil's pressure on the walls, then sized each wall as a flat plate. With the top edge free, every candidate material bowed past the 2 mm limit. A 2 × 2 × ¼ in aluminum tube rim braces that edge and brings all seven under it, even with the test plate driven 5 mm from a wall.



Peak wall deflection for each candidate material, without and with the top rim



Deflection map of a 1 in acrylic wall: 4.56 mm peak at the free top edge with no brace (left), 1.63 mm with the top rim (right)

How I got there: soil-stress and lateral earth-pressure model for the compacted simulant; load-path analysis through the actuator, mounts and fasteners, including the off-center plate case; plate-bending sizing of seven wall materials; and weighted decision matrices for wall material, frame and compaction method. The team reproduces all of it in a Python notebook as its reference.

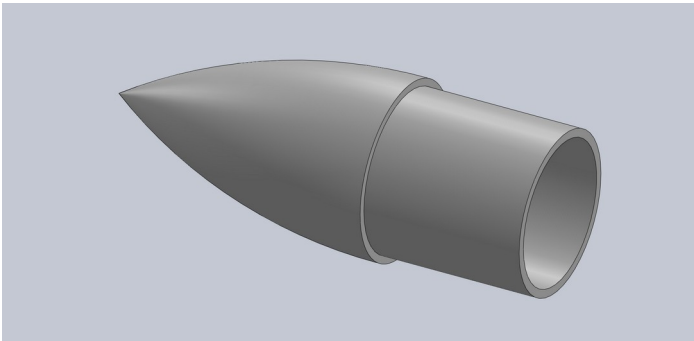
Rocket Club — Tripoli Level 1 Certification

Colorado School of Mines Rocket Club • Spring 2026 • Designer and builder

I designed the rocket in OpenRocket, modeled and 3D-printed its nose cone, laser-cut the fins and built the full vehicle on a tight launch timeline. It flew on an AeroTech H motor with a clean parachute recovery, earning Tripoli Level 1 high-power certification.



On the pad, liftoff, and the Level 1 certification pin



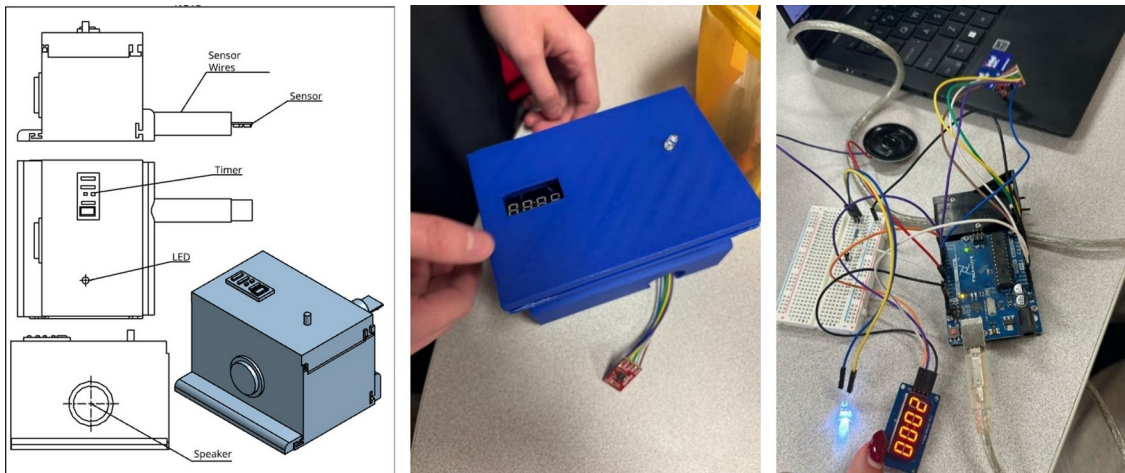
Nose cone designed in SolidWorks and printed in one piece on a Bambu X1 Carbon

Spec	Value
Vehicle	83 mm diameter, about 1 m tall, 38 mm motor mount
Build	Cardboard airframe, laser-cut plywood fins, 3D-printed PLA ogive nose cone
Simulated flight (H219T)	649 m (2,128 ft) apogee, Mach 0.49
Recovery	36 in parachute, about 6 m/s landing

Tourniquet Feedback Device

Red Rocks Community College • Spring 2025 • Team lead, 5 students

Most tourniquet failures happen because the user can't tell when it is tight enough. Our clip-on device reads blood flow below the tourniquet and, once flow stops, lights an LED, sounds an alarm and starts a timer for the next level of care. I led the team and the stakeholder work with EMS and military medics, fabricated about 75% of the hardware and wrote about 25% of the software.



Enclosure design, the printed enclosure, and all subsystems running on the bench

Result	Value
Validation tests	8 of 8 subjects classified correctly (4 people, 4 objects)
Civilian confidence in tourniquet placement	More than doubled with the device
Hardware	Pulse-oximeter sensor, Arduino, LED, speaker, timer, 3D-printed enclosure
Projected unit price	About \$133